CHAEHYUN LIM

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AEROSPACE SYSTEMS | HARDWARE INTEGRATION, TEST & ELECTRIFICATION

Georgia Tech Aerospace Engineering B.S. graduate and M.S. candidate with hands-on experience in instrumented test automation, Li-polymer battery-pack testing, hybrid-electric propulsion architecture, and control-system prototyping. Combines hardware/software troubleshooting with mission-energy modeling and multidisciplinary interface coordination.

EDUCATION

Georgia Institute of Technology / *M.S. Aerospace Engineering* | Atlanta, GA — Expected May 2027

Georgia Institute of Technology / *B.S. Aerospace Engineering* | GPA 3.65/4.00, **Highest Honor** — May 2026

Coursework: Control Systems Analysis & Design · Circuits & Electronics · Instrumentation & Electronics Lab · Aerospace Systems Engineering (current) · Design & Analysis of Experiments (current)

ENGINEERING EXPERIENCE & PROJECTS

Moore Dynamics & Analytics Lab / *Undergraduate Researcher* | Atlanta, GA — Oct 2024 – May 2026

- Built MATLAB/Arduino C++ automation for stepper-driven impact testing, integrating NI cDAQ force/acceleration acquisition, calibration, response visualization, and data logging.
- Resolved driver-mismatch and MATLAB global-variable faults; analyzed repeated impact responses and designed SolidWorks fixtures, adjusting Formlabs SLA print orientation, supports, and resin settings.
- Implemented camera-pattern recognition and target-coordinate calculation; used the UR30 teach pendant for robot positioning during final tests.

43rd VFS Student Design Competition / *Propulsion Lead* | Atlanta, GA — Aug 2025 – Jun 2026

- Developed a conceptual hybrid-electric tiltrotor power architecture and MATLAB mission-energy models to assess turbogenerator/battery power allocation across hover, conversion, and cruise.
- Compared turbogenerator and motor candidates by power-to-weight, efficiency, practicality, and technology readiness; tracked sizing iterations against RFP priorities and physical feasibility.
- Advanced from Deputy Team Lead to Propulsion Lead; coordinated power, mass, and packaging interfaces across a 12-person team. Team placed **2nd globally** in the undergraduate division.

CubeSat Reaction-Wheel Heading Control / *Senior Capstone* | Atlanta, GA — Jan – May 2026

- Designed and bench-validated a single-axis PI-II reaction-wheel prototype; selected components, created the KiCad schematic, and assembled and troubleshot Arduino/IMU/ESC/BLDC hardware.
- Wrote most of the final report; team test results: 2.7 s rise, 5.9 s settling, <8.24% overshoot, ±2.8° heading error. Ranked #1 in class; **Top 5** capstone project.

Vertical Lift Research Center of Excellence / *Undergraduate Researcher / Test Operator* | Atlanta, GA — Jan – May 2025

- Performed 6S Li-polymer battery-pack cycle-life tests per a published ≤1C charge / 750 W constant-power discharge protocol, following LiPo safety procedures and maintaining test records.
- Collected pre-/post-discharge resistance measurements and pack-temperature data; reviewed capacity degradation and subsequently analyzed internal-resistance results in Excel.

Forest Fire Detection UAV / *Pilot & System Designer* | Atlanta, GA — Aug 2024 – Dec 2025

- Integrated the airframe, a 3D-printed battery enclosure, and thermal/optical payloads; performed preflight checks and pre-programmed waypoint flights to verify vehicle stability and payload function.

TECHNICAL LEADERSHIP & OPERATIONS

Korea Defense Intelligence Command / *Sr. Squad Leader / Intelligence Editor in Chief* | South Korea — Oct 2020 – Apr 2022

- Led an 11-person squad in 24/7 four-shift operations; authored a 100-page operating guide and redesigned workflow, increasing team daily processing from ~15–20 to 30–40 items.
- Oversaw 33 defense/aerospace publications; selected as 1 of 4 interpreters for the 2021 U.S.–ROK combined exercise. **Recognized by the Chairman of the ROK Joint Chiefs of Staff**.

TECHNICAL SKILLS & CERTIFICATIONS

Test & Instrumentation: NI cDAQ · Force/acceleration sensing · Oscilloscope · Multimeter · Bench power supply · Function generator

Software & Data: MATLAB/Simulink · Arduino C++ (AccelStepper) · Excel · Python (coursework)

CAD & Prototyping: SolidWorks · KiCad (schematic capture) · SLA 3D printing · Soldering/wiring · UR30 (teach pendant)

Pilot Credentials: FAA Part 107 · EASA A1/A3 · ROK UAV Pilot · 20+ logged UAV flight hours

Languages: Korean (native) · English (fluent) · Chinese (HSK 5)